Home Assistant integration -- pending proposal

Status: proposed; not implemented or tested. This page turns the need for the assistant to understand current home state without sending every sensor into model context into a project requirement.

Problem

A Home Assistant installation may contain many sensors and entities. Adding every state to each message consumes context, increases cost/latency, and can hide the information needed to answer the question. Readings may also be stale, unavailable, or incomplete.

Desired outcome

When asked about the home, Vladmir should query only the data needed for that question and produce a concise, verifiable, current answer. For example, a question about the living room should query relevant entities in that area rather than attaching the entire inventory.

Proposed architecture

1. Explicit local connector: operator-configured connection to a selected Home Assistant instance. Use a supported official API (REST or WebSocket, to be confirmed during implementation) and store its token outside the repository, in the system secret store when available or a local file with restrictive permissions.
2. Lightweight catalog: load areas, names, domains, and entity IDs; do not send every sensor value to the model by default.
3. On-demand query: map the question to filters by area, entity type, name, and time period. Fetch only the current states needed and query history only when a trend or comparison is requested.
4. Compact summary: group results by room/category, discard attributes unrelated to the question, and enforce item/token limits. A short-lived cache may reduce calls; every displayed datum must retain its read timestamp.
5. Traceable answer: state in natural language which areas/entities were queried, the timestamp, and whether an entity was unavailable or stale. Do not claim the state is complete when a query fails.
6. Separate actions: first version is read-only. Any service that changes state (lights, locks, climate, alarm, etc.) must require a specific permission and user confirmation; a question about status does not authorize an action.

Privacy and security

- Disabled by default; no automatic discovery or cloud data transfer.
- Restrict connection to the explicitly configured endpoint. Since Home Assistant is commonly on a private network, implement a dedicated connector with allowlisting and destination validation; do not weaken SSRF protections in general web browsing.

- Keep the token out of Git, prompts, logs, URLs, and error messages.
- Expose separate least-privilege scopes: inventory, current state, and history. Do not request
  control access for the read-only version.
- Treat entity names, states, and attributes as untrusted data, not instructions to the model or
  tools.
- Bound entity count, attribute size, history window, timeout, query frequency, and concurrent
  requests.
- Allow the operator to remove configuration, cache, and associated local data.

Acceptance criteria

- [ ] The user can configure/disable the local integration; the secret is not versioned or
  exposed in logs.
- [ ] A room query fetches only relevant entities, and the model request contains a bounded
  summary rather than the full inventory.
- [ ] A trend query uses a bounded history window and reports the period used.
- [ ] The answer identifies timestamps, unavailable/stale states, and integration failures
  without inventing readings.
- [ ] Entity names/attributes that look like instructions cannot redirect assistant behavior or
  trigger tools.
- [ ] Tests cover timeout, invalid token, removed entity, unavailable value, oversized payload,
  malformed response, and disallowed destination attempts.
- [ ] The first version does not execute control services. If control is added later, each
  operation has its own scope, clear preview, and confirmation.
- [ ] Integration testing uses a lab instance and synthetic entities before residential use.

Suggested phases

1. Investigate supported Home Assistant API versions and choose REST or WebSocket based on
   pagination, history, and events.
2. Implement a read-only connector and configuration/diagnostics UI.
3. Create a lightweight local catalog and selective query by area/domain.
4. Add bounded history, timestamps, and stale-data indicators.
5. Evaluate token budget, latency, and accuracy with synthetic scenarios containing many
   entities.
6. Discuss device automation only after permission and confirmation review.

Outside this proposal

- This is not a feature currently available in Vladmir.
- It does not imply unrestricted local-network access or device discovery.
- It does not send the entire home state to the model in every conversation.
- It does not enable home control in the first version.
- It does not promise compatibility with every Home Assistant installation, add-on, integration,
  or entity.